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Article

Two-Dimensional Metal–Organic Framework with High-Performance Single-Molecule Magnets as Nodes Showing Magnetic Coercivity Photomodulation

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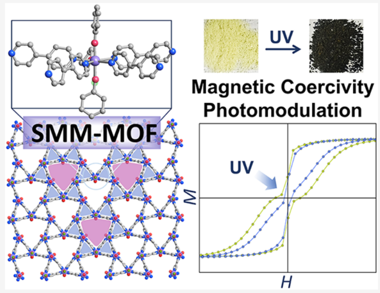
ABSTRACT: Addressing the spatial organization of high-performance single-molecule magnets (SMMs) and achieving stimuli-responsive switching of their magnetic bistability are pivotal challenges in molecular memory technologies, paving the way for advanced opto-magnetic devices. Herein, we utilize the photosensitive ligand 4,4′-bipyridine (BPy) as a linker to incorporate typical pentagonal-bipyramidal SMMs as nodes into a two-dimensional metal-organic framework (MOF), formulated as $\{[Dy_{1.5}(OPh)_2Cl(BPy)_3(THF)_{1.5}][(BPh_4)_{1.5}]\cdot 0.5THF\}_n$ (**1**). The precise synthesis facilitates axial coordination of PhO^- and equatorial alignment of BPy, enforcing perpendicular orientations of the principal magnetic axes of Dy^{3+} ions across all Kagomé layers. Compound **1** exhibits photochromic behavior upon exposure to ultraviolet irradiation at room temperature, driven by a photoinduced electron transfer process that generates radicals. The resulting **1uv** displays overall faster relaxation dynamics compared to **1**, characterized by shorter relaxation times at identical temperatures within the 12–70 K range, a lower diverging temperature in field-cooled and zero-field-cooled curves (9 K for **1** vs. 6 K for **1uv**), and reduced energy barriers from 1048(17)/822(46) K for **1** to 1000(9)/641(34) K for **1uv**. Notably, the coercive field decreases dramatically from 4500 Oe for **1** to 1300 Oe for **1uv** at 2 K, while the hysteresis loop opening temperature decreases from 20 K for **1** to 14 K for **1uv**. These photoinduced changes are due to the formation of photogenerated radicals and alterations in crystal packing. This work achieves an MOF that integrate high-performance SMM behavior with magnetic coercive photomodulation, providing a design paradigm for engineering advanced SMM-MOFs with tailored photomagnetic switching.



■ INTRODUCTION

Single-molecule magnets (SMMs)—discrete molecules that relax magnetization slowly and exhibit magnetic hysteresis as well as quantum effect at low temperatures—are prime candidates for high-density information storage and quantum computing.[1−8] Recently, lanthanide SMMs have witnessed remarkable progress, including energy barriers over 1000 K, and magnetic hysteresis above liquid nitrogen temperature.[9−16] These advances are predominantly driven by two molecular platforms: sandwich-type complexes with conjugated aromatic ligands and pentagonal/hexagonal bipyramidal coordination systems, where strategic ligand field engineering enables strong axial magnetic anisotropy. To realize SMM-based memory applications, two critical issues require resolution: individual molecule spin-state detection and scalable integration strategies that preserve magnetic functionality in densely packed architectures. Metal-organic frameworks (MOFs) offer distinct advantages in this context, as their periodic structures can enable precise spatial organization of SMM building blocks, tunable inter-spin distances and maintained magnetic isolation, thereby facilitating the creation of dense, addressable spin arrays.[17,18] The reported SMM-MOF strategies mainly involve either direct incorporation of SMMs at MOF nodes or encapsulation of SMMs within MOF channels.[18−27] Nevertheless, the development of MOFs incorporating SMMs with energy barriers approaching 1000 K remains unexplored, due to the difficulty in maintaining the chemical integrity of prefabricated SMMs during framework assembly. More critically, achieving a unified alignment of magnetic anisotropy across metal ions within an SMM-MOF presents a significant hurdle.

The precise control of magnetic states via external stimuli, such as electric potential, pressure, solvent, temperature, or light, broadens the scope of practical applications for SMMs.[27−31] In particular, photo-switchable SMMs enable the use of light as a highly effective stimulus, offering high spatial and temporal resolution. To date, only a small number of Ln-SMMs exhibiting photo-responsive magnetism have been reported.[32−49] Despite this progress, significant challenges remain, as these complexes face multiple substantial limitations: (1) Light-induced ligand reactions, such as ring-opening/closing, cis/trans isomerization, and cycloaddition, often occur far from the first coordination

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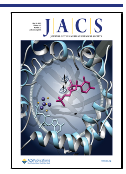

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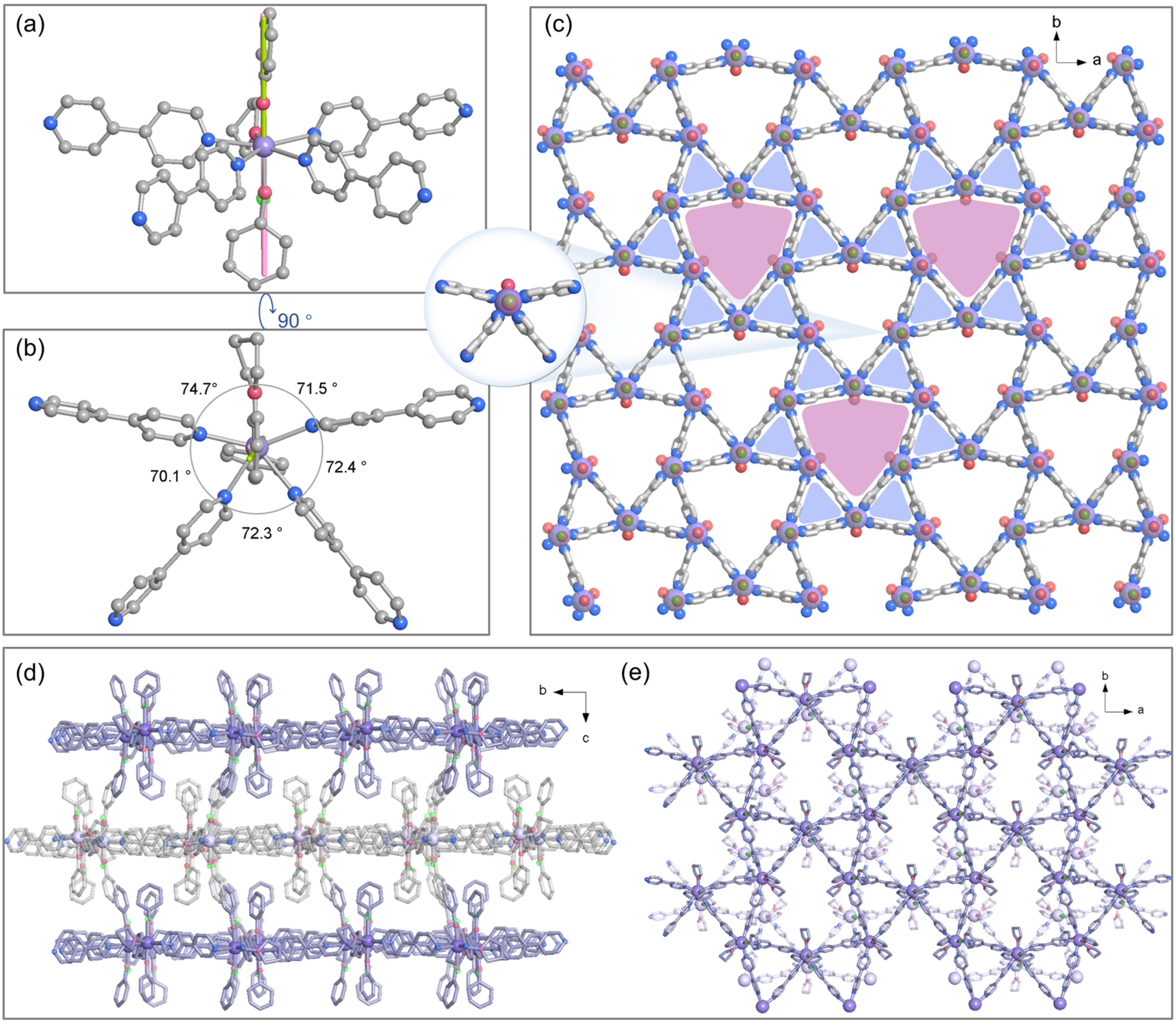

Figure 1. Structure of the $[Dy(OPh)X(BPy)_4(THF)]^+$ ($X = Cl/PhO^-$) cation in **1** viewed from side (a) and from above (b). The pink and green solid lines represent *ab initio* calculated orientation of the main magnetic axis on Dy1 in the ground Kramer's doublets for $[Dy(OPh)_2(BPy)_4(THF)]^+$ and $[Dy(OPh)Cl(BPy)_4(THF)]^+$, respectively. (c) Kagomé layer of **1**. Colors: O = red, C = grey, N = blue, Dy = purple, Cl = green. BPh_4^- anions, one disordered PhO^- moiety and hydrogen atoms are omitted for clarity. (d) AB hierarchically packing mode of Kagomé layers. (e) The crystal structure of **1**.

sphere of lanthanide ions, resulting in weak or negligible photomagnetic effects.[32−44] Similar issues exist in the few reported photoinduced electron transfer (PET) systems.[49] (2) The energy barriers of known photo-responsive Ln-SMMs typically do not exceed 500 K.[41,46] (3) Magnetic hysteresis, a prerequisite for data storage, is often absent or limited to butterfly-like loops that lack coercive fields and remnant magnetization.[38−41] Very recently, Tong's group reported a Dy-SMM functionalized with a photochromic polyoxomolybdate anion, exhibiting ultraviolet-driven modulation of magnetic hysteresis.[46]

Here, we present an effective strategy for constructing two-dimensional MOFs (2D MOFs) that combine high-performance SMM behavior with an efficient photomagnetic response. By employing an axial strong-field phenoxide ligand and equatorial photosensitive 4,4′-bipyridine (BPy) ligand, we isolated three 2D MOFs $\{[RE_{1.5}(OPh)_2Cl(BPy)_3(THF)_{1.5}][(BPh_4)_{1.5}]\cdot 0.5THF\}_n$ (RE = Dy (**1**), Y (**2**) and Gd (**3**)). Compound **1** features a pentagonal-bipyramidal Dy-based SMM at its nodes, displaying an energy barrier for magnetization reversal of 1048(17)/822(46) K and a coercive field of 4500 Oe at 2 K. Interestingly, the magnetic principal axes of Dy^{3+} ions are well-aligned in a perpendicular orientation across all layers. Furthermore, this SMM-MOF shows photochromic behavior at room temperature under ultraviolet (UV) irradiation, accompanied by a pronounced UV-light-actuated photomagnetic effect and photomodulation of magnetic coercivity.

RESULTS

Synthesis and Structure of the SMM-MOF. Compounds **1**−**3** were synthesized in a one-pot reaction in THF, beginning with the established synthesis step of $[RE(OPh)Cl(THF)_5]^+$ (RE = Dy, Y, Gd),[50] followed by the addition of PhPy (4-phenylpyridine), and finally introducing the BPy ligand (see the

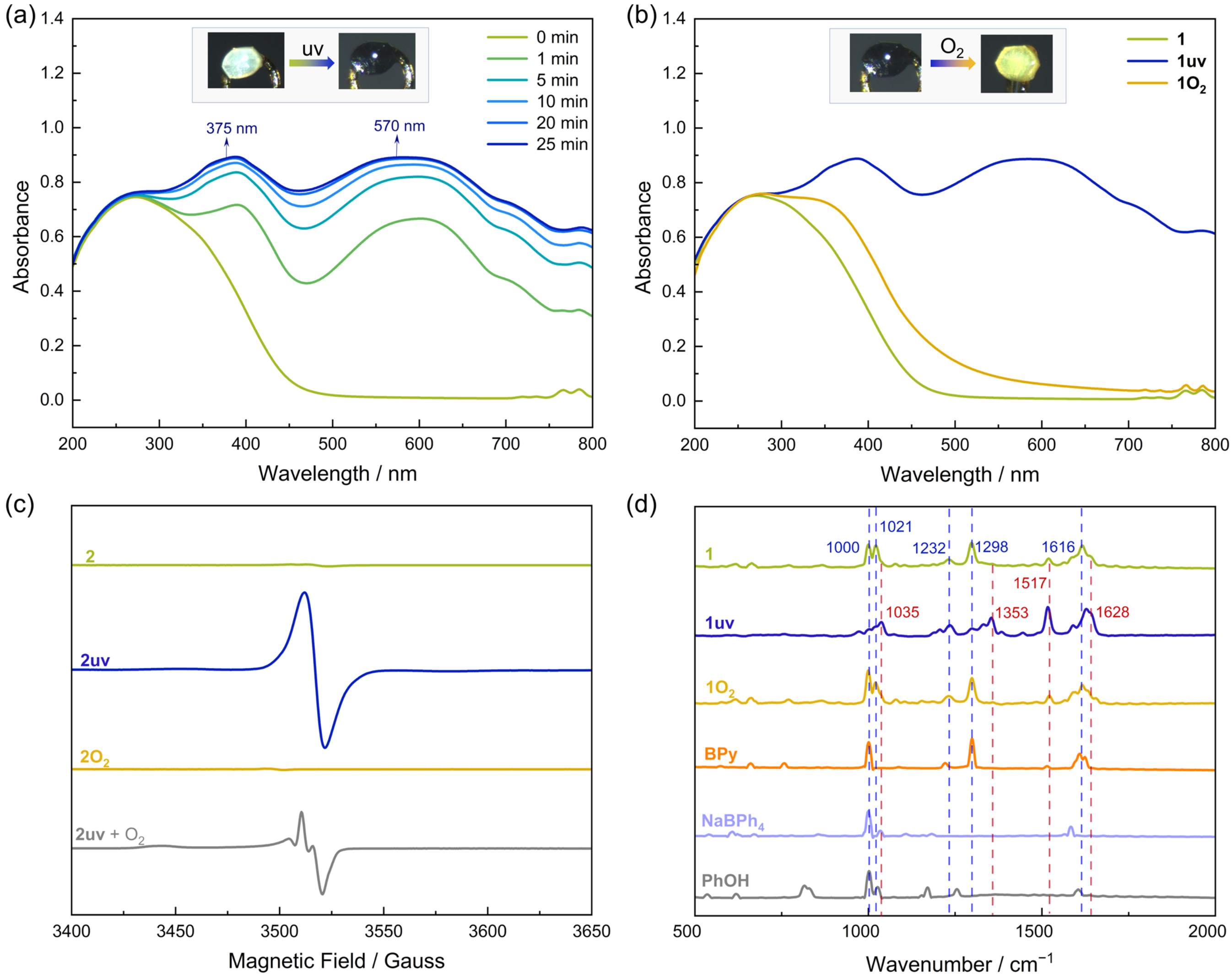

Figure 2. (a) Time-dependent solid-state UV−vis spectra of **1** upon UV irradiation (365 nm). Insert: the crystal forms of **1** and **1uv**. (b) Solid-state UV−vis spectra of **1uv** decolorization under O_2 oxidation. Insert: the crystal forms of **1uv** and **1O$_2$**. (c) EPR spectra for **2**, **2uv**, and **2O$_2$** measured in argon at room temperature, with **2uv** + O_2 being tested directly after thorough exposure to O_2. (d) Raman spectra of **1**, **1uv**, **1O$_2$**, PhOH, $NaBPh_4$, and BPy. Excitation wavelength: 785 nm.

Supporting Information, Figures S1 and S2, and Table S1). The key challenge in the synthesis lies in the effective coordination of the PhPy ligand, as its presence is essential for the successful formation of **1**−**3**. We attempted to grow single crystals of the intermediate after the addition of PhPy, ultimately succeeding by slightly modifying the reaction ratio. The resulting structure $[Dy(OPh)_2(PhPy)_5](BPh_4)$·5Benzene (**4**) (Figure S3 and Tables S2 and S3), combined with the previously reported structures $[Dy(O^tBu)_2(PhPy)_4]^+$ and $[Dy(O^tBu)_2(PhPy)_5]^+$ demonstrates that the intermediate is crucial for the subsequent substitution by BPy, likely due to a lowered activation energy.[51,52]

Single-crystal X-ray diffraction reveals that **1** crystallizes in the monoclinic space group $C2/c$ (Table S4). The asymmetric unit consists of one and a half crystallographically independent Dy^{3+} ions, two $PhO^−$, one $Cl^−$, three BPy, along with THF molecules and $BPh_4^−$ counterions (Figure S4). Both Dy^{3+} ions (Dy1 and Dy2) adopt a similar pentagonal-bipyramidal configuration, with axial coordination by $PhO^−/Cl^−$ ligands (albeit disordered) and equatorial coordination by four BPy ligands and one THF molecule (Figures 1a, b and S5). Thus, they can be fully described as $[Dy(OPh)_2(BPy)_4(THF)]^+$ (**1a**) and $[Dy(OPh)Cl(BPy)_4(THF)]^+$ (**1b**) species, with a ratio of 1:2 (vide infra). The bond lengths of Dy−$O_{(PhO)}$ range from 2.034(13) to 2.12(4) Å, while the Dy−Cl bond lengths span 2.499(14) to 2.533(16) Å, consistent with those reported for $[Dy(OPh)_2(THF)_5]^+$, $[Dy(OPh)_2(Py)_5]^+$ and $[Dy(OPh)Cl(THF)_5]^+$ (Table S5).[50] In comparison, the equatorial Dy−$O_{(THF)}$ bond lengths (2.337(13)−2.393(15) Å) and Dy−$N_{(BPy)}$ bond lengths (2.501(10)−2.553(9) Å) are significantly longer. The axial $O_{(PhO)}$−Dy−$O_{(PhO)}$ and $O_{(PhO)}$−Dy−Cl angles range from 172.7(11)° to 178.3(3)°, while all adjacent equatorial bond angles fall within 70.1(4) to 74.7(4)°, closely approximating to the ideal values for a pentagonal-bipyramidal geometry. This geometry (D_{5h}) can also be confirmed by the SHAPE values (1.010, 0.733 and 0.605) (Table S8).[53]

Each Dy(III) ion coordinates to four BPy ligands, and each BPy ligand bridges two Dy(III) ions, forming a twisted Kagomé layer (Figure 1c). Each triangle within the layer is equilateral, consisting of $Dy_3(BPy)_3$ units with internal angles of 60°. Meanwhile, each hexagon adopts an irregular, shield-like shape, composed of $Dy_6(BPy)_6$ units with internal angles of 158° and

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82°. Such Archimedean tessellation is labeled as $(3.6)^2$ (Figure S6). The shortest Dy···Dy distance within the layer is about 12.1 Å.

The Kagomé layers stack in an AB mode, where triangular and hexagonal motifs exhibit a staggered conformation across neighboring layers (Figure 1d,e). The centroids of the hexagons in one layer align almost vertically above those of the hexagons in the subjacent layer (offset = 2.3 Å). Consequently, this architecture preserves hexagonal channels. The charge-balancing BPh_4^- anions are embedded between the Kagomé layers, with two-thirds located in triangular cavities of adjacent layers (Figures S7 and S8). The remaining third of anions occupy the large hexagonal channels but are highly disordered relative to the symmetry elements due to their weak interactions with the framework. Detailed analysis of interlayer forces at known locations shows that each benzene ring of BPh_4^- and certain hydrogen atoms on BPy/PhO^- generate C−H···π interactions, with distances of 2.4−2.9 Å (hydrogen atom to ring centroid) (Figure S9). The interlayer spacing is 11.0 Å.

The Y(III) analogue **2** and Gd(III) analogue **3** possess crystal cell parameters very similar to those of **1** (Table S10). However, **2** displays X-ray photochromism when exposed to Cu Kα radiation for extended periods due to its weak diffraction (Figure S10). Meanwhile, **3**, which suffers from severe polycrystallinity, also undergoes a subtle color change upon prolonged exposure to Mo Kα radiation. Importantly, their unit cell parameters remain unchanged after the color change. Furthermore, scanning electron microscopy (SEM) images reveal that the dried crystals exhibit a layered stacking structure, which explains the strong polycrystallinity (Figure S11).

Photochromic Properties. In an inert atmosphere, the yellowish powder of **1** undergoes a rapid color change to deep blue-violet (denoted as **1uv**) upon 365 nm (6 W) ultraviolet irradiation (Figure S12). This deep blue-violet sample exhibits exceptional stability, maintaining its state for months when stored in a glove box in the dark at ambient temperature. Exposure to O_2 quickly reverts the powder to the yellow form (denoted as **1O$_2$**), whereas visible light and high temperatures fail to trigger this oxidation process. **2** and **3** also display reversible photochromic behavior, with the corresponding states labeled **2uv**, **2O$_2$**, **3uv** and **3O$_2$**, respectively.

This photochromic transition is also observed in single crystals of **1**, **2** and **3** (Figures 2 and S13). Taking **1** as an example, a remarkable rapid response to UV light in air is observed in single-crystal X-ray diffraction experiments. The light-yellow crystals of **1**, coated with perfluoropolyether, quickly turn deep blue-violet within seconds under 365 nm UV light (6 W) in air. The crystals revert to yellow within seconds after UV irradiation ceases in air. The unit cell parameters of **1**, **1uv**, and **1O$_2$** remain nearly unchanged (Table S4). Similar behavior is observed for **2**, **2uv**, **2O$_2$**, **3**, **3uv**, and **3O$_2$** (Table S11).

To explore the photochromic behavior, solid-state ultraviolet-visible (UV−vis) spectra were measured. The pristine sample of **1** displays a strong absorption peak near 265 nm, primarily attributed to the $\pi-\pi^*$ transition of the BPy ligand (Figure 2a). Additionally, two weak peaks around 780 nm correspond to the f−f electronic transitions of Dy^{3+} ions.[54,55] Upon irradiation with 365 nm UV light (6 W), two new electron absorption bands emerge at around 375 and 570 nm. These bands are assigned to the electronic transitions of the BPy anion radical, indicating the single-electron reduction of BPy.[56−58] The peak intensity increases with prolonged irradiation time, reaching saturation after 25 min. When **1uv** exposed to O_2, the UV−vis absorption peaks at around 375 and 570 nm nearly disappear within 30 min. However, **1uv** does not fully revert to the original state of **1** (Figure 2b). Crystallographic cell parameters, powder X-ray diffraction (PXRD), and infrared (IR) spectroscopy confirm the structural integrity throughout the coloring−decoloring process (Table S4, Figures S14 and S15). The photochromic activation−deactivation of **2** and **3** are also validated by UV−vis spectra and PXRD (Figures S14, S16 and S17).

It is noteworthy that **1**−**3** also undergo slow discoloration under visible light, indicating broader photochromic activity. Exposure of **1** to ambient light for 3 days results in a color change to light blue, accompanied by a weak electronic absorption band at 570 nm (Figure S18). A similar discoloration and electronic absorption band are observed upon irradiation with 520 nm green light (7 W) (Figure S19).

Solid-state electron paramagnetic resonance (EPR) spectroscopy performed in an argon atmosphere confirms the presence of radicals. A sharp, strong signal for **1uv** is detected at a frequency of 9.86 GHz with g = 2.0000, while no such signal is observed for **1** and **1O$_2$** (Figure S20). For the diamagnetic Y(III)-based **2**, a very weak radical signal at a g-factor of 2.0040 is observed, resulting from a minimal quantity of light-induced contamination (Figure 2c). After irradiation, this EPR signal for **2uv** is significantly enhanced, suggesting the generation of radicals and electron transfer to the BPy ligand. Notably, the EPR signal of **2uv** exposed to O_2 reveals the presence of superoxide anion radicals.[59,60] Following full decolorization, pump/fill cycles, which effectively exclude interference from superoxide anion radicals, render EPR signals for **2O$_2$** nearly absent, demonstrating that the radicals are almost entirely quenched by O_2. For **3**, **3uv** and **3O$_2$**, the EPR signals are nearly identical, with a g-value of 1.9800 (Figure S21). This is due to the pronounced signal from Gd(III), resulting from its half-filled f^7 electron configuration, which obscures the sharper radical signals.

The Raman spectra were further studied (Figure 2d). Before irradiation, the band at 1616 cm^{-1} for **1** corresponds to the ring stretching vibration of BPy, while the band at 1298 cm^{-1} represents the C−C stretching between the pyridine rings. Additionally, the peaks at 1000 and 1021 cm^{-1} are attributed to the ring breathing modes of the pyridine and benzene rings.[57,61,62] After irradiation, the vibrational band originally positioned at 1616 cm^{-1} upshifts to 1628 cm^{-1} for **1uv**, accompanied by the emergence of a prominent band at 1517 cm^{-1}, indicating charge transfer to BPy, which affects the stretching vibration of the pyridine ring. The 1353 cm^{-1} band exhibits a 55 cm^{-1} upshift relative to the 1298 cm^{-1} band for **1**, attributed to the formation of a quinone-like structure and the reduction of BPy aromaticity.[61,62] Meanwhile, the band at 1000 cm^{-1} is apparently diminished for **1uv**, while a new band at 1035 cm^{-1} emerges as a strong feature, consistent with observations in reported viologen compounds and bipyridine radicals.[58,61,62] After further oxidation, **1** and **1O$_2$** exhibit nearly identical peak positions. However, the peak at 1021 cm^{-1} does not fully recover, suggesting partial structural reversibility, as corroborated by the UV−vis spectra. Similar Raman spectral changes are observed for the Y(III) and Gd(III) analogues (Figures S22 and S23).

To assess the aging effect, five consecutive photochromic cycle tests were performed on **1**. It maintains photochromic responsiveness through five activation-inactivation cycles. However, the persistence of the coloring and decoloring

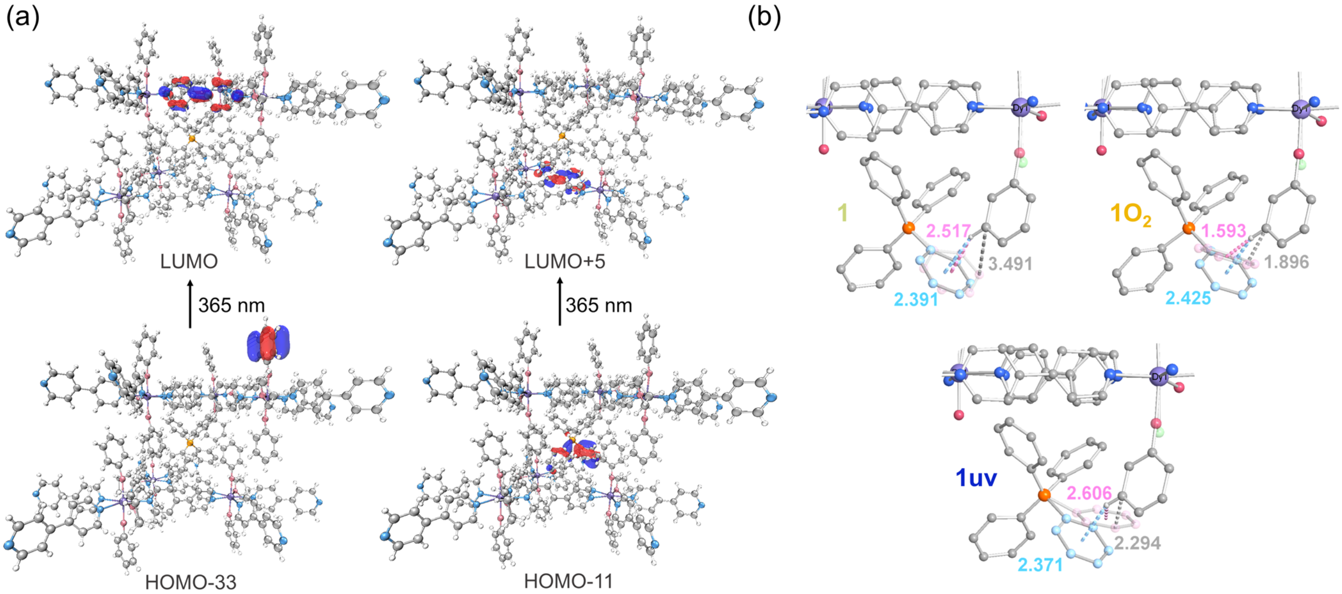

Figure 3. (a) DFT-calculated the possible electron transition process in compound **1** when irradiated at 365 nm with an extended model of **1A**. (b) The distance variation between one disordered phenyl ring of the BPh_4^- anion and a neighboring PhO^- ring in **1**, **1uv**, and **1O$_2$**.

processes is limited. As the cycles progress, the color of the colored samples shifts from deep blue-violet to light blue, while the decolored samples change from light yellow to deep yellow. Five-cycle UV−vis spectra validate this incompletely reversible process (Figure S24). The yield of photogenerated BPy radicals decreases with cycling, with significant fatigue observed after the second irradiation cycle. Additionally, the peak at 1035 cm^{-1} in the Raman spectra of the decolored samples gradually intensifies after repeated cycles, suggesting progressive structural changes in the material (Figure S25).

PET Mechanism. Insights into the mechanism of photoinduced electron transfer (PET) are derived from density functional theory (DFT) calculations (Supporting Information). For computational modeling of **1**, two extended crystal structures based on $[Dy(OPh)_2(BPy)_4(THF)]^+$ and $[Dy(OPh)Cl(BPy)_4(THF)]^+$ were selected as model systems (denoted as **1A** and **1B**, respectively). Inspection of the orbitals close to the lowest unoccupied molecular orbital (LUMO) indicates that the electron densities are predominantly localized on the BPy groups (Tables S12−S15). For the orbitals near the highest occupied molecular orbital (HOMO), the densities are primarily concentrated on the PhO^- and BPh_4^- groups for **1A**, and on the PhO^-, BPh_4^-, and Cl^- groups for **1B**. The calculated energy gaps between these HOMOs and LUMOs correspond to wavelength ranges of 141−618 nm for **1A**, 146−515 nm for **1B**, in excellent agreement with the experimentally observed photoresponse under UV−visible light irradiation. Among these transitions, those corresponding to the experimental wavelength of 365 nm mainly involve electron transfer from electron-rich PhO^- and BPh_4^- to electron-deficient BPy, as transitions from Cl^- to BPy require higher energy. The potential electron transition pathways under 365 nm irradiation are depicted in Figures 3a and S26.

The single-crystal structures of **1**, **1uv**, and **1O$_2$** were determined at 150 K using the same crystal (Tables S4−S7). Although polycrystallinity and disorder in the crystal data complicate the analysis of minute bond changes on the picometer scale relevant to PET processes,[49,63] a significant structural alteration is observed in a disordered phenyl ring of the BPh_4^- anion. This particular phenyl ring approaches a neighboring PhO^- ring, with the shortest C···C distance between the two rings decreasing from 3.491 Å in **1** to 2.294 Å in **1uv**, and further to 1.896 Å in **1O$_2$** (Figure 3b). Such variation offers strong evidence for enhanced intermolecular forces. Therefore, the PET process is suggested to proceed through the following mechanism: under 365 nm irradiation, photoinduced electron transfer from BPh_4^- and PhO^- to BPy ligands generates $BPh_4^\bullet$, $PhO^\bullet$ radicals, and $BPy^{\bullet-}$ radical anions. This irradiation leads to a reduction in the distance between one disordered phenyl ring of the BPh_4^- anion and one neighboring PhO^- ring, suggesting an interaction or potential pairing effect between $BPh_4^\bullet$ and $PhO^\bullet$ radicals. During the oxidation step, $BPy^{\bullet-}$ transfers an electron to O_2, reducing O_2 to $O_2^{\bullet-}$. Meanwhile, the closer proximity of $BPh_4^\bullet$ and $PhO^\bullet$ radicals, combined with the disappearance of the EPR signal in **1O$_2$**, further suggests a localization or pairing effect of these radicals. Therefore, BPy can oscillate between two redox states, whereas BPh_4^- cannot. With each of the five irradiation cycles, new electrons are supplied to sustain the continuous generation of photogenerated radicals. However, the gradual depletion of electron donors leads to a decline in radical generation efficiency, resulting in the observed fatigue behavior.

Photomagnetic Properties. The magnetic properties of **1** and **1uv** were studied through direct current (dc) and alternating current (ac) magnetic susceptibility measurements. At 300 K, the $\chi_M T$ value of **1** (20.57 cm^3 K mol^{-1}) is close to the theoretical value of 21.26 cm^3 K mol^{-1} for 1.5 free Dy(III) ions (J = 15/2 and g = 4/3) (Figure S27). Upon cooling, the $\chi_M T$ value decreases slowly from 300 K to around 16 K, and then drops rapidly to 9.45 cm^3 K mol^{-1} at 2 K. The field-dependent magnetization reaches a maximum of 7.22 Nβ at 2 K and 70 kOe (Figure S28). The unsaturated magnetization and the non-superposed M vs. H plots indicate the presence of magnetic anisotropy and/or low-lying excited states.

1 was irradiated for 30 min until fully converted to the deep blue-violet form **1uv**, and magnetic measurements were then conducted under the same conditions. The $\chi_M T$ trend for **1uv** closely resembles that of **1**, with a slow decrease followed by a

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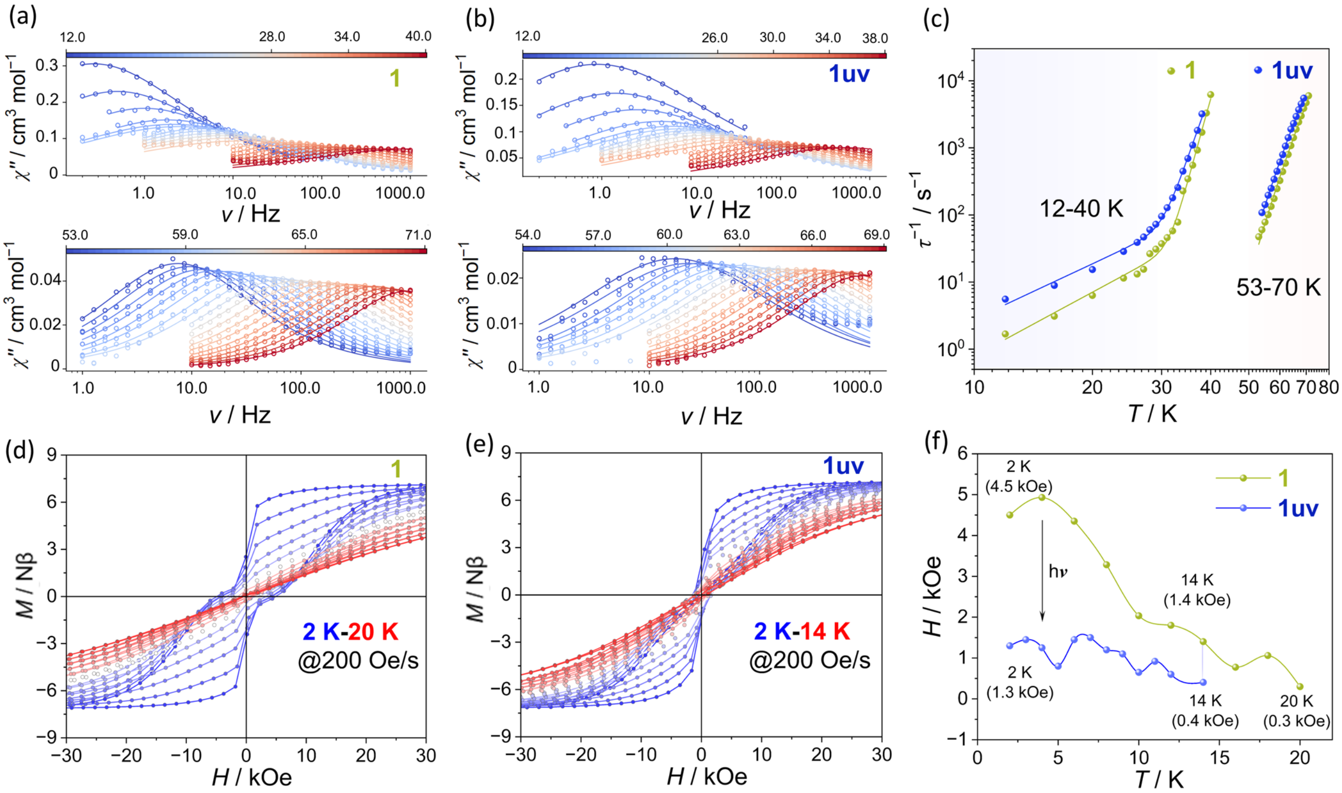

Figure 4. (a) Frequency-dependence of the out-of-phase (χ'') ac susceptibility for **1** from 12.0 to 40.0 K and from 53.0 to 71.0 K under zero dc field. (b) Frequency-dependence of the out-of-phase (χ'') ac susceptibility for **1uv** from 12.0 to 38.0 K and from 54.0 to 69.0 K under zero dc field. (c) Plots of the magnetic relaxation rate vs. temperature for **1** and **1uv**. The solid lines are best fits. Hysteresis loops for **1** (d) and **1uv** (e) at various temperature at an average sweep rate of 200 Oe/s. (f) The coercive field (H_c) vs. temperature (T) for **1** and **1uv**.

sharp drop as the temperature decreases (Figure S27). Key differences include: (1) a slightly higher $\chi_M T$ value for **1uv** at 300 K (20.63 cm^3 K mol^{-1}), potentially due to the generation of a small amount of radicals; and (2) a higher $\chi_M T$ value for **1uv** at 2 K (12.75 cm^3 K mol^{-1}), indicating significant differences in low-temperature magnetic properties. **1uv** also exhibits unsaturated magnetization (7.31 Nβ at 2 K and 70 kOe) and non-superposition of M vs. H plots, with a slight difference in maximum magnetization relative to **1** (Figure S29).

Below 71 K, the in-phase (χ') and out-of-phase (χ'') susceptibilities of **1** and **1uv** display obvious temperature- and frequency-dependent behavior, characteristic of slow relaxation of magnetization (Figures 4a,b and S30–S33). Both compounds show two distinct relaxation regimes: a high-temperature domain (53–71 K, referred to as relaxation domain #1) and a low-temperature domain (12–40 K, referred to as relaxation domain #2). The two relaxation domains cannot be simultaneously defined within the 1–1000 Hz owing to their disparate time scales. Therefore, the characteristic relaxation time (τ) at each temperature was extracted by globally fitting the frequency-dependent χ' and χ'' data to the generalized Debye model using the CC-FIT2 program (Figures S32–S35).[64,65] The obtained α value roughly increases as the temperature decreases ranging from 0.14 to 0.44 for **1** and 0.20 to 0.53 for **1uv** (Tables S16 and S17). The high α values ($\geq$0.40) observed at low temperatures suggest a wide distribution of the relaxation times and the coexistence of multiple relaxation pathways. The τ^{-1} vs. T plot on a $\log_{10}$–$\log_{10}$ scale is employed to analyze the dynamic relaxation pathways. Fitting relaxation domain #1 with the Orbach process $\tau^{-1} = \tau_0^{-1}\, e^{-(U_{eff})/T}$ provides an effective energy barrier of U_{eff} = 1048(17) K and τ_0 = 7.0(19) × 10^{-11} s for **1**, and similarly, U_{eff} = 1000(9) K and τ_0 = 9.2(13) × 10^{-11} s for **1uv**. For relaxation domain #2, fitting with a combination of Raman and Orbach processes using the equation $\tau^{-1} = \tau_0^{-1}\, e^{-(U_{eff})/T} + CT^n$ produces U_{eff} = 822(46) K, τ_0 = 2.2(27) × 10^{-13} s, C = 5.5(35) × 10^{-4} s^{-1} K^{-n} and n = 3.2(2) for **1**, and U_{eff} = 641(34) K, τ_0 = 1.8(17) × 10^{-11} s, C = 5.1(24) × 10^{-3} s^{-1} K^{-n} and n = 2.7(2) for **1uv** (Figure 4c). In the 12–70 K range, **1uv** exhibits shorter relaxation times compared to **1** at identical temperatures (Figure S36, Tables S16 and S17).

The field-cooled (FC) and zero-field cooled (ZFC) magnetization curves measured at 1000 Oe reveal divergence for both **1** and **1uv**, indicating a possible transition from the paramagnetic state to long-range ordering, a superparamagnetic state, or a spin glass state (Figures S37 and S38).[66] Notably, **1** displays superior magnetic characteristics, with a higher irreversibility temperature (T_{IRREV} = 9 K) compared to **1uv** (T_{IRREV} = 6 K). Furthermore, **1** shows a distinct ZFC peak maximum at 2000 Oe (corresponding to blocking temperature T_b = 5 K), while this feature remains absent in **1uv**, implying three plausible mechanisms in the latter: stronger quantum tunneling of magnetization (QTM), reduced blocking temperature, or a broader relaxation time distribution (as indicated by larger α values, Tables S16 and S17 and Figures S39 and S40).[5]

Magnetic hysteresis measurements on polycrystalline **1** reveal hysteresis loops up to 20 K at a scan rate of 200 Oe s^{-1}, with a coercive field (H_c) of 4500 Oe and remanent magnetization (M_r) of ~2.52 Nβ at 2 K (Figure 4d). In contrast, **1uv** exhibits narrowed hysteresis loops with a H_c of 1300 Oe at 2 K, decreasing to nearly zero at 14 K, at which H_c for **1** remains at

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1400 Oe (Figure 4e). This also enables the coercive field to be photo-switched off within the 14−20 K temperature range (Figure 4f). The reduction in coercive field for **1uv** demonstrates irradiation-enhanced QTM, as confirmed by abrupt hysteresis collapse near zero field, low-temperature ac susceptibility tail, and the aforementioned FC-ZFC characteristics. In a recent study, Tong's group demonstrated optical control of the coercive field in a Dy-SMM.[46] To our knowledge, these two studies are among the rare example of the practical realization of UV-light-actuated magnetic coercivity photomodulation in Ln-SMMs.

The magnetic properties of $\mathbf{1O_2}$ were further explored. The $\chi_M T-T$ curve shows a trend similar to **1** and **1uv**, with $\chi_M T$ values of 20.51 $cm^3\ K\ mol^{-1}$ at 300 K and 10.54 $cm^3\ K\ mol^{-1}$ at 2 K (Figure S27). At 2 K and 70 kOe, the magnetization reaches 7.11 Nβ (Figure S41). The FC and ZFC magnetization curves diverge at 8 K at 1000 Oe (Figure S42), and $1O_2$ exhibits double relaxation behavior analogous to **1** and **1uv** (Figure S43). For $\mathbf{1O_2}$, the coercive field and remanent magnetization at 2 K are approximately 1500 Oe and 2.57 Nβ, respectively (Figure S44). This aligns with previous findings, as oxygen scavenges photogenerated free radicals, hindering the sample from returning to the original state.

To explore the generation of free radicals and their influence on magnetic exchange interactions, the dc magnetic susceptibility of diamagnetic Y(III)-based and isotropic Gd(III)-based compounds were investigated under identical conditions. For **2uv**, the $\chi_M T$ value at 300 K is 0.09 $cm^3\ K\ mol^{-1}$, combined with a maximum magnetization of 0.12 Nβ at 2 K and 70 kOe, reflecting a relatively low radical yield of approximately 10% (Figures S45 and S46). Interestingly, this approximate value closely matches the number of phenyl ring from BPh_4^- that approach PhO^- during the structural transformation from **1** to **1uv**. For **3** and **3uv**, the nearly identical $\chi_M T-T$ curves as well as $M-H$ plots at various temperatures further support a low radical yield and slight change in exchange interactions after irradiation (Figures S47 and S48). Curie-Weiss analysis provides Weiss constants of $\theta = -2.19$ K (**3**) and −2.29 K (**3uv**), corresponding to weak antiferromagnetic interactions ($J = -0.10$ K for **3** and −0.11 K for **3uv**) (Figures S49 and S50). Given the low radical yields in certain photochromic compounds and the extremely small Dy···Dy distance contraction ($\Delta d \approx 0.005$ Å for **1uv**), it is expected that the magnetic interactions in **1** and **1uv** differ only marginally (Figure S51).[67−69]

To elucidate the magnetic relaxation mechanism, we performed *ab initio* calculations using the CASSCF/RASSI-SO/SINGLE−ANISO approach implemented in the OpenMolcas program.[70−72] Despite the 2D nature of **1**, the local coordination environment surrounding each Dy(III) ion plays a pivotal role in determining the crystal field splitting and energy barrier. Herein, the crystal structures $[Dy(OPh)_2(BPy)_4(THF)]^+$ (**1a**-Dy1) and $[Dy(OPh)Cl(BPy)_4(THF)]^+$ (denoted as **1b**-Dy1 for Dy1, and **1b**-Dy2 for Dy2) were selected as models (Figures S52−S54), retaining their complete coordination environments without optimization. For **1a**-Dy1, **1b**-Dy1 and **1b**-Dy2, the ground-state doublet is predominately characterized by the m_J = |±15/2> wave function, with the principal magnetic axis aligned along the pseudo-C_5 axis (Tables S18−S23). The first excited state for all three models is almost pure m_J = |±13/2> state, with the g_z value closely aligned to the pseudo-C_5 axis. For **1b**-Dy1 and **1b**-Dy2, the second excited state exhibits a highly mixed wave function, with the largest component being 35 % |±3/2> for **1b**-Dy1 and 43 % |±1/2> for **1b**-Dy2, resulting in a g_z value nearly perpendicular to the pseudo-C_5 axis. In contrast, this mixed state occurs only in the third excited state for **1a**-Dy1, while the second excited state remains a pure m_J = |±11/2> state. Therefore, the magnetic relaxation likely proceeds via the third excited state, with a predicted U_{eff} of 1312 K for **1a**-Dy1, and via the second excited state, with U_{eff} of 760 K for **1b**-Dy1 and 644 K for **1b**-Dy2 (Figures S55−S57).

For **1uv**, the computational models employed the same structural motifs: $[Dy(OPh)_2(BPy)_4(THF)]^+$ (**1c**-Dy1) and $[Dy(OPh)Cl(BPy)_4(THF)]^+$ (denoted as **1d**-Dy1 for Dy1 and **1d**-Dy2 for Dy2), with two key approximations: exclusion of radical spin contribution and idealized charge state assignments (Figures S58−S60). Similar to **1**, the relaxation of **1c**-Dy1 occurs via the third excited state with U_{eff} of 1325 K. The **1d**-Dy1 and **1d**-Dy2 proceed through the second excited state, with U_{eff} of 857 and 652 K, respectively (Tables S24−S29 and Figures S61−S63). These results demonstrate that structurally analogous units in **1** and **1uv** exhibit virtually identical magnetic relaxation pathways, along with similar energy level structures, consistent with their minimal structural variations.

The above experimental and computational results suggest several key points:

(1) The unique architecture of this SMM-MOF facilitates an ordered arrangement of SMMs, with each SMM's principal magnetic easy axis oriented perpendicular to the layer plane. This perpendicular alignment is uniformly maintained across all stacked layers.

(2) The Orbach relaxation at higher temperatures is contributed by $[Dy(OPh)_2(BPy)_4(THF)]^+$, while the process at lower temperatures is attributed to $[Dy(OPh)Cl(BPy)_4(THF)]^+$, consistent with reports for compounds possessing similar axial coordination environments.[50] This observation reinforces the critical role of axial ligands in determining energy barriers of pentagonal-bipyramidal SMMs. Similarly, while the reference complex $[DyCl_2(THF)_5]^+$ with the same axial ligands exhibits relaxation behavior below 10 K (U_{eff} = 83 K),[73] no corresponding feature is observed in **1**. Quantitative single-crystal XRD refinement further limits any $[DyCl_2(BPy)_4(THF)]^+$ content ≤ 8%. These results indicate that $[DyCl_2(BPy)_4(THF)]^+$ is either absent or present in undetectable trace amounts.

(3) Compared to **1**, **1uv** displays accelerated relaxation dynamics throughout the entire temperature range, characterized by shorter relaxation times at identical temperatures within the 12−70 K range, a lower FC-ZFC divergence temperature (6 K for **1uv** vs. 9 K for **1**), reduced energy barriers (1000(9)/641(34) K for **1uv** vs. 1048(17)/822(46) K for **1**), a lower blocking hysteresis temperature (14 K for **1uv** vs. 20 K for **1**) and narrowed hysteresis loops (e.g., 1300 Oe for **1uv** vs. 4500 Oe for **1** at 2 K).

(4) For the changes in dynamics governed by the Orbach process, the coordination of photogenerated $BPy^{\bullet-}$ and $PhO^{\bullet}$ radicals to the Dy(III) center perturbs the primary coordination sphere (∼10% radical generation). A close inspection of the coordination environment reveals a reduction in axial linearity (O−Dy−O: −1.4°; O−Dy−Cl: −0.6-2.1°), and non-systematic variations in bond lengths (Figure S64). Continuous Shape Measures (CShM) analysis reveals the Dy(III) center in **1uv**

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displays increased deviation from ideal pentagonal-bipyramidal geometry (D_{5h} symmetry), with SHAPE values 14−31% higher than in **1** (Tables S8 and S9). This geometric distortion, coupled with an altered electron density distribution, collectively decreases the energy barriers.

For the significantly enhanced QTM at low temperatures under zero field after irradiation, several structural modifications are crucial. Although the radical-induced Dy···Dy distance contraction is minimal (0.005 Å), consistent with only slightly enhanced antiferromagnetic interactions in the Gd analogues, its impact at cryogenic temperatures could be considerable. Detailed analysis of the crystal packing reveals more pronounced intermolecular distance contraction, particularly as phenyl rings from BPh_4^- approach PhO^- during the structural transformation from **1** to **1uv**. Closer packing likely enhances spin-phonon coupling and promotes QTM efficiency.[39,74,75] Furthermore, **1uv** exhibits increased average M−L bond length variation (0.0145 Å for **1** vs 0.0156 Å for **1uv**), indicating enhanced ligand vibrations and reduced molecular rigidity. Such changes in local vibrational modes may facilitate stronger QTM effects, ultimately leading to the observed reduction in coercive fields.[76]

■ CONCLUSIONS

We present an efficient approach to integrate pentagonal-bipyramidal SMMs with high magnetic anisotropy into a 2D MOF using photoactive BPy linkers. The resulting SMM-MOF achieves a record energy barrier exceeding 1000 K and a coercive field of 4500 Oe at 2 K. It exhibits photochromic behavior at room temperature upon UV irradiation, accompanied by accelerated relaxation dynamics, as demonstrated by shorter relaxation times and narrowed hysteresis loops. These changes are likely due to photogenerated radicals that alter the crystal field, exchange interactions, and crystal packing. To our knowledge, this system represents a rare example of UV-light-actuated photomodulation of magnetic coercivity in Ln-SMMs, offering a new platform for the design of optically responsive magnetic materials with molecular-scale ordered arrays.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c03704.

Experimental details for synthesis, single-crystal XRD study; powder XRD study; UV−vis spectra; IR spectra; EPR spectra; Raman spectra; SEM images; XPS data; magnetic measurements; and theoretical calculation (PDF)

Accession Codes

Deposition Numbers 2371969 and 2427861−2427864 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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